



Scoreboard

Parallel ATA Hard Drives

BRAND	Samsung	Western Digital	Hitachi Deskstar	Western Digital	Western Digital	Hitachi Deskstar
MODEL	SPO411N	WD800BB	HD5728080PLAT20	WD1200BB	WD1600BB	HDT722525DLAT80
Features (Out of 15)	4.99	5.48	4.38	5.97	6.46	6.43
HDD Capacity (Formatted in GB)	37.27	74.53	76.69	111.79	149.05	232.88
Interface	Parallel ATA	Parallel ATA	Parallel ATA	Parallel ATA	Parallel ATA	Parallel ATA
Rated Capacity (GB)	40	80	80	120	160	250
Rotational Speed (RPM)	7200	7200	7200	7200	7200	7200
Buffer Size (MB)	2	2	8	2	2	8
Jumper If Required (bundled)	✓/✓	✓/✓	✓/✓	✓/✓	✓/✓	✓/✓
Bundled Cables/Screws	✓/✓	✓/✓	✓/✗	✓/✓	✓/✓	✓/✗
Performance (Out of 70)	37.76	40.71	41.10	39.30	39.36	46.04
Synthetic Tests						
HD Tach 3.0.1.0						
8 MB Sample File						
Average Read (MBps)	49.4	51.9	49	47.6	51.2	56.2
Average Write (MBps)	42.5	46.9	44.4	46.6	50.2	49.7
CPU Utilisation (%)	1	1	2	1	1	2
Random Access Time (ms)	14.1	13.5	13.4	13	18.1	12.6
Read Burst Speed (MBps)	68.7	95.7	122.9	93.8	93.7	128.7
32 MB sample file						
Average Read (MBps)	48.3	51.9	49	52.7	57.2	56.1
Average Write (MBps)	48.3	50.5	47.7	51	55.1	54.1
CPU Utilisation (%)	2	1	1	2	1	2
Random Access Time (ms)	14.2	13.5	13	13	18	12.5
Read Burst Speed (MBps)	68.9	96	114.9	93.8	93.7	128.6
SiSoft Sandra 2007 Engineers Edition						
Drive Index (MBps)	46	50	50	49	53	58
Sequential Read (MBps)	54	56	58	58	63	67
Random Read (MBps)	34	39	41	38	40	46
Sequential Write (MBps)	54	56	58	58	63	66
Random Write (MBps)	36	39	32	29	30	36
Access Time (ms)	11	8	7	9	9	7
Real World Tests In Seconds (lower is better)						
File Write Test 4 GB Assorted Files	145.6	75.8	143.2	142.3	140.4	114.3
File Read Test 4 GB Assorted Files	117	75	119.5	125.3	117.7	111.2
Internal File Read Test 4 GB Assorted Files	294.5	300.6	422.3	345.4	362	208.7
File Write Test 4 GB Single File	78.5	142.1	76	74.1	69.3	70.6
File Read Test 4 GB Single File	78	74.4	74.5	72.3	69.1	70.4
Internal File Read Test 4 GB Single File	263.5	317	309.3	361.5	364.6	192.3
Photoshop CS2 Startup Time	12.2	10.5	11.4	10.6	10.4	8.7
Photoshop CS2 Image Load (550 MB file)	27.3	24.8	26.2	25.9	24.7	22
Photoshop CS2 Image Load (1 GB file)	85.2	77.1	66.5	80.98	76.8	64.2
Farrry (level 1) load time	26	30.6	18.2	21.9	30	23.2
Warranty (Years)	5	3	5	3	3	5
Price Per MB (Paise)	5.24	3.34	3.82	3.06	2.75	2.39
Price Index (Out of 15)	13.50	10.59	9.00	7.71	6.43	4.74
Price (Rs)	2,000	2,550	3,000	3,500	4,200	5,700
Total Score (Out of 100)	56.25	56.77	54.48	52.98	52.25	57.21



+ Decent performance with larger files, slim design
- Limited storage



+ Decent performance for its price
- Limited storage



+ Performed well in single file-copy tests
- Cheaper options available in 80 GB segment



+ Good when working with large files
- 2 MB buffer



+ Low CPU utilisation, speedy performer
- Lack of 8 MB buffer



+ Fastest of the PATA drives, good for graphic applications
- Costly